

Capstone Project Status Report #1

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Our project will be a web grocery store program named UltraMart where users can purchase from a variety of items and add them to their cart. Users will be able to create their own accounts, which involve a username and a password. The website will authenticate accounts using a database containing usernames and passwords. Users will also be able to view their profile and add a small description and change the name that publicly appears. Each grocery item will have its own price, description, name, and stock count. If an item is out of stock, the user will not be able to purchase it. For transactions between users, they can send friend requests to each other. This allows users to view what other people have recently purchased. In terms of messaging, users can send messages to their friends. Each message will be stored in the database.

The UltraMart project will satisfy requirements by being an online web application designed to optimize online grocery shopping and account management. Being an online grocery shopping application, it will address and handle authentication and user account management while also providing various ways to interface with site functions. Through tools within Java and our numerous databases there will be mechanisms such as dashboards for incorporating messaging and transaction monitoring and databases for managing associated information will be employed. Using the project requirements as a guide, we aim to employ the group work practices we have learned to create not only a functional online grocery application, but an efficient and well designed tool that addresses a real world situation that is universally understood.

UltraMart follows a three-tier MVC (Model-View-Controller) architecture that will separate the web application into a frontend, backend, database layer that will be hosted on Hostinger. The frontend is built with HTML, CSS, and Javascript. It will run directly in the

user's browser and handle all user-facing pages such as the store, shopping cart, user profile, and messaging interface. Javascript will be used to make REST API calls to the backend and fetch/send data dynamically. The backend is a JAVA application server hosted on Hostinger. It exposes a REST API that the frontend communicates with over HTTP using JSON. The backend has 4 core responsibilities: authentication of account management(login, registration, session handling) , store/cart logic(item listings, stock management, purchases) , and social features(friend requests, messaging, and purchase feeds). The database uses MYSQL, hosted on Hostinger. The backend will communicate with MYSQL using JDBC(Java Database Connectivity). The database stores all persistent data; user accounts, grocery items, stock counts, messages, and order/transaction history. This architecture was chosen for its simplicity, clear separation of concerns, and compatibility with Hostinger's hosting environment. It is well-documented and appropriate for the scope of the capstone project.

For the first milestone, the responsibilities of each team member will involve setting up the basics of the website. Dan will be the project manager, and will work on the frontend of the website, which includes things such as the CSS and layout of the website. Thomas and Moua will work on the backend, including things such as the Java code and the database tables. The goal for the first milestone is to set up a basic website with at least two webpages that shows how we would like to lay out the information.

Since our last meeting, we have decided on a website host and finalized the project description for our grocery store website. Additionally, we have organized our roles in the project and decided on the languages we will use. For our next meeting, more concrete details about the website itself will be established. This involves the layout of the pages and how information will be conveyed to the user. We will create diagrams to help us visualize the layout,

too. We will utilize Github during these next two weeks, and aim to create a database for the website.

Our group plans to use Github to upload code such as HTML pages, CSS pages, and Javascript files. This will make it easier for us to view each other's code before uploading it to the web host. Additionally, Hostinger allows shared access among multiple users, which means each group member can upload files and access things such as the database. VSCode will be used for the IDE.

All team members develop on Windows machines. For code editors and IDE's, the team uses a mixture of Visual Studio Code and IntelliJ IDEA. VS Code is the preferred for frontend work given its strong HTML/CSS, and JavaScript support. The backend will be worked around a mixture of VS Code and IntelliJ using the support from those IDE's. Version Control is managed through GitHub, using a main/dev branching development strategy. Developers work on the dev branch and merge into main at stable milestones. Code sharing is handled through pull requests and pushes, with peer code reviews between team members to maintain code quality and shared understanding of the codebase. For team communication, the group uses Google Meet for meetings and Google Chat for quick day-to-day communication. In terms of skill development distribution, the team is comfortable working on the backend side, with strengths in JAVA and MySQL. Dan takes the lead on the frontend, handling HTML, CSS, and Javascript. While Thomas and Moua focus primarily on the backend logic database management. This distribution aligns well with the project's three-tier architecture. The team uses a manual testing approach and peer code reviews to help strengthen debugging skills and give each member hands on experience troubleshooting a full application stack.

The project timeline will be based on the milestones and status reports as we plan on reflecting and reassessing with each milestone to ensure that our development is going according to plan and that we are remaining on track with following and satisfying the requirements.

Milestone 1, we plan to have all of our tools for collaboration created and accessible by all, such as github, hostinger, and google docs. As well as creating the framework for our databases and the project architecture within the site hosting service. Finally, with all of the tools and initial framework laid out, we will divide the work-load then combine our contributions and refine prior to the next milestone.